
ISO 26262 Part 3 - Item Definition

System: Wiper System

Generated on: 2025-10-08 10:14:08

Work Product: 5.5.1 – Item definition resulting from requirements in 5.4

Generated: 2025-10-08 10:12:30

1. Introduction

Reference: 5.1, 5.2

The Wiper System, as per ISO 26262-3:2018, is a vehicle function responsible for clearing the windshield. Its primary function is to ensure driver visibility by removing rain, snow, and debris, and its scope encompasses the entire mechanism from the motor to the wiper blades. This document's purpose is to analyze its hazards and risks (HARA) and define safety requirements for its functional safety.

2. Overview

Reference: 5.4.1

2.1 Vehicle Communication Architecture

Reference: 5.4.2 d), f)

- **Communication Protocols:** CAN, LIN
- **Primary ECUs:** Body Control Module (BCM), Wiper Control Module (WCM)
- **Gateway Role:** BCM often acts as a gateway, translating messages between the Wiper System and other vehicle networks (e.g., Powertrain CAN).

2.2 Vehicle Relevant Information

Reference: 5.4.1 d)

Here are some vehicle class, propulsion type, and key performance targets relevant to a wiper system:

- **Vehicle Class:** M1 (Passenger Cars), N1 (Light Commercial Vehicles)
- **Propulsion Type:** BEV (Battery Electric Vehicle), ICE (Internal Combustion Engine)

***Key Performance Targets:*

- **Wipe Speed/Frequency:** Ability to clear precipitation effectively across a range of speeds (e.g., 30-60 wipes per minute).
- **Durability/Lifespan:** Expected operational cycles before component failure (e.g., 1 million cycles).

2.3 Vehicle State Machine

Reference: 5.4.1 b)

Here's how the Wiper System's behavior changes based on key vehicle states:

- **Off:** The wiper system is inactive. It will not operate unless explicitly activated by the user, even if rain is detected. This state prioritizes power saving and prevents accidental operation.
- **Drive (and Ignition On):** The wiper system is active and responsive to user input (manual control) and potentially automatic rain sensing. It can be set to various speeds or modes (intermittent, low, high) to clear the windshield for visibility.
- **Charge (EVs):** While the vehicle is charging, the wiper system's behavior is generally similar to "Drive" if the ignition is on or in an accessory mode. However, some systems might have a "standby" mode where wipers are only activated by direct user input to conserve power during charging, especially if the vehicle is parked and not in use.

2.4 Item Expected Operation

Reference: 5.4.1 b), 5.4.2 f)

Here's a concise description of a typical operational profile:

- **Duty Cycle:** Intermittent, with periods of high activity followed by idle or low-power states.
- **Lifetime:** 5-10 years or 100,000-200,000 km, depending on usage and maintenance.

-
- **Environmental Conditions:** Operates in a wide range of temperatures (-20°C to 60°C), humidity (10-90% non-condensing), and may encounter dust, vibration, and moderate shock.

2.5 Law Specifications and Homologation Criteria

Reference: 5.4.1 a)

Here are key legal regulations directly impacting 'Wiper Systems', with their relevance:

***UNECE Regulation No. 16 (Safety belts and restraint systems):*

- While not directly about wipers, this regulation's focus on driver visibility and control indirectly influences wiper system performance requirements to ensure safe operation.
- It emphasizes the importance of unobstructed vision for safe driving, which wipers are critical for maintaining.

***UNECE Regulation No. 48 (Installation of lighting and light-signalling devices):*

- This regulation mandates that vehicles must have adequate lighting, and by extension, the ability to maintain clear visibility through the windshield, which wipers facilitate.
- It indirectly supports the need for effective wiper systems to ensure drivers can see and be seen, especially in adverse weather conditions.

***UNECE Regulation No. 107 (Retrofitting of passenger cars with regard to the installation of lighting and light-signalling devices):*

- This regulation, concerning retrofitting, highlights the importance of maintaining original safety equipment, including functional wiper systems, for continued roadworthiness.
- It underscores that modifications should not compromise essential safety features like clear visibility, which wipers provide.

***National Regulations (e.g., FMVSS in the US, ECE in Europe, ADR in Australia):*

- These overarching national standards often incorporate or reference specific performance requirements for wiper systems to ensure they effectively clear the windshield for safe driving.
 - They typically mandate minimum performance levels for wiper speed, coverage, and durability under various environmental conditions.
-

2.6 Quality Constraints

Reference: 5.4.1 c)

Here are a few critical quality targets, quantified where possible:

- **Mean Time Between Failures (MTBF):** > 20,000 hours (for critical electronic components).
- **Certification:** IATF 16949 compliance (for automotive supply chain quality management).

3. Item Description

Reference: 5.4.2

3.1 Item Architecture

Reference: 5.4.2 a), e)

Wiper System - System-Level Description (ISO 26262-3 Clause 5.4.2)

The Wiper System is a functional unit responsible for clearing the vehicle's windshield to ensure driver visibility. It operates autonomously based on environmental conditions and driver input.

(1) Main Physical Components:

- **Wiper Motor:** An electric motor that drives the wiper arms.
- **Wiper Arms:** Mechanical linkages that hold and move the wiper blades across the windshield.
- **Wiper Blades:** Rubber or silicone elements attached to the wiper arms that make contact with the windshield to remove water and debris.
- **Wiper Control Module (ECU):** The central processing unit that manages the operation of the wiper system.
- **Park Position Sensor:** Detects when the wiper arms are in their designated resting position.
- **Wiper Speed Control Resistor/Module (if applicable):** Provides different operational speeds for the wiper motor.

(2) External Interfaces:

- **Vehicle Power Supply:** Receives electrical power to operate the components.
-

-
- **CAN Bus:** For communication with other vehicle ECUs, such as the Body Control Module (BCM) or Powertrain Control Module (PCM).
 - **LIN Bus (or dedicated wiring):** For communication with specific sensors, such as the rain sensor.

(3) User Interfaces:

- **Wiper Control Switch (stalk on steering column):** Allows the driver to select wiper speed (e.g., off, intermittent, low, high), activate the washer function, and potentially control intermittent delay.
- **Washer Fluid Pump Activation:** Triggered by the wiper control switch to spray washer fluid onto the windshield.
- **Indicator Light (optional):** May illuminate on the dashboard to indicate a system fault or specific operational status.

(4) Key Interactions with Other Vehicle Systems:

- **Receives rain intensity from rain sensor via LIN:** The rain sensor provides information about the amount of precipitation on the windshield, influencing the automatic operation of the wipers.
- **Receives vehicle speed from ABS/ESP module via CAN bus:** Vehicle speed may be used to adjust wiper speed (e.g., increase speed at higher vehicle velocities).
- **Receives ignition status from BCM via CAN bus:** The wiper system is enabled or disabled based on the vehicle's ignition state.
- **Communicates wiper status (e.g., operating, parked, fault) to BCM via CAN bus:** The BCM may use this information for dashboard indicators or other vehicle functions.
- **Receives washer fluid level status from BCM via CAN bus (if applicable):** Informs the driver or system about the washer fluid reservoir level.

3.2 Block Diagram

Reference: 5.4.2 a)

Wiper System Block Diagram: Textual Description

This document provides a textual description of a typical Wiper System block diagram, outlining its main components, internal and external interfaces, and sensors/actuators. The structure is designed for clarity and understanding of the system's functionality.

1. Main System Components/Modules:

The Wiper System is comprised of several key functional blocks that work in concert to operate the windshield wipers. These are:

- **Wiper Control Module (WCM):** This is the central processing unit of the wiper system. It receives inputs from various sensors and user controls, processes this information, and generates commands for the wiper motor and other actuators. It manages different wiper speeds, intermittent operation, and park functionality.
- **Wiper Motor:** This is the primary actuator responsible for physically moving the wiper arms across the windshield. It receives power and control signals from the WCM to operate at different speeds and directions.
- **Wiper Linkage Mechanism:** This mechanical assembly connects the wiper motor to the wiper arms. It translates the rotational motion of the motor into the oscillating motion required for the wipers.
- **Wiper Arms & Blades:** These are the visible components that sweep the windshield. The arms are attached to the linkage mechanism, and the blades are attached to the arms, providing the wiping action.
- **Power Distribution Module (PDM) / Fuse Box:** This module provides the necessary electrical power to the wiper system components. It includes fuses and relays to protect the system from overcurrents and to control power flow.

2. Internal Interfaces:

These interfaces represent the communication pathways and connections *between* the main system components/modules.

- **WCM to Wiper Motor Control Interface:** This is a communication and power interface. The WCM sends control signals (e.g., speed commands, direction, on/off) and potentially power modulation signals to the Wiper Motor. The Wiper Motor may also send feedback signals (e.g., motor speed, position) back to the WCM.
 - **Wiper Motor to Linkage Interface:** This is a purely mechanical interface. The output shaft of the Wiper Motor connects directly to the Wiper Linkage Mechanism, transferring rotational force.
 - **Linkage to Wiper Arms Interface:** This is also a mechanical interface. The Wiper Linkage Mechanism has connection points for the Wiper Arms, allowing the oscillating motion to be transmitted.
-

-
- **WCM to PDM/Fuse Box Interface:** This is an electrical interface. The WCM receives power from the PDM/Fuse Box. It may also send signals to control relays within the PDM/Fuse Box (e.g., to activate the wiper motor circuit).
 - **PDM/Fuse Box to Wiper Motor Power Interface:** This is a direct electrical power supply line from the PDM/Fuse Box to the Wiper Motor, providing the necessary voltage and current for operation.

3. External Interfaces:

These interfaces represent the connections of the Wiper System to the *outside world* or other vehicle systems.

- **User Input Interface (Wiper Stalk/Switch):** This is the primary human-machine interface. The driver interacts with the Wiper Stalk (typically on the steering column) to select wiper speeds (off, intermittent, low, high), activate the washer fluid, and potentially control other wiper functions. This interface sends user commands to the WCM.
- **Vehicle Power Supply Interface:** The entire Wiper System draws its primary electrical power from the vehicle's main battery and charging system, managed through the PDM/Fuse Box.
- **Body Control Module (BCM) / CAN Bus Interface (Optional but common):** In modern vehicles, the WCM may communicate with the BCM or other ECUs via a vehicle network (e.g., CAN bus). This allows for integration with other vehicle functions like rain sensing, automatic headlights, or diagnostic reporting. This interface would carry data such as rain sensor readings, vehicle speed, or diagnostic trouble codes.
- **Washer Fluid Pump Control Interface:** The WCM typically controls the washer fluid pump. This interface sends a signal from the WCM to activate the pump when the washer function is selected by the user.

4. Sensors/Actuators:

This section lists the specific sensors and actuators within or connected to the Wiper System.

Sensors:

-
- **Wiper Motor Position Sensor (Optional):** Some systems include a sensor to provide feedback on the exact position of the wiper motor or arms. This is crucial for accurate "park" functionality and potentially for advanced features.
 - **Rain Sensor (Optional, often external to WCM):** This sensor detects the presence and intensity of rain on the windshield. It typically sends a signal to the WCM to enable automatic or intermittent wiper operation.
 - **Vehicle Speed Sensor (Indirect input via BCM/CAN Bus):** While not directly part of the wiper system, vehicle speed can influence wiper operation (e.g., increasing speed at higher vehicle speeds). This information is usually obtained via the vehicle's network.

Actuators:

- **Wiper Motor:** As described in the main components, this is the primary actuator that drives the wiper arms.
- **Washer Fluid Pump:** This actuator is responsible for spraying washer fluid onto the windshield. It is controlled by the WCM.
- **Wiper Motor Relay (within PDM/Fuse Box):** This is an electrically controlled switch that connects or disconnects power to the wiper motor, managed by the WCM.

This textual description provides a comprehensive overview of the Wiper System's architecture, enabling a clear understanding of its constituent parts, their interactions, and its overall functionality.

3.3 Item Components

Reference: 5.4.2 a)

Let's break down the core physical and logical components of a typical automotive Wiper System.

Wiper System Components:

Here's a breakdown of the main physical and logical components, focusing on the core elements:

1. Wiper Control Module (WCM) / Wiper ECU

-
- **Component Name:** Wiper Control Module (WCM) or Wiper Electronic Control Unit (ECU)
 - **Type:** ECU (Electronic Control Unit)
 - **Main Function:** This is the "brain" of the wiper system. It receives inputs from various sensors and user commands, processes this information, and then sends control signals to the wiper motor and other actuators. Its core functions include:
 - **Intermittent Wipe Control:** Managing the timing and duration of wiper sweeps in intermittent mode based on speed and potentially rain sensor input.
 - **Wiper Speed Control:** Adjusting the speed of the wiper motor (low, high, and potentially variable speeds).
 - **Park Position Control:** Ensuring the wipers return to their designated "park" position at the base of the windshield when turned off.
 - **Washer Pump Activation:** Triggering the washer pump when the washer function is selected.
 - **Diagnostics:** Monitoring system health and reporting faults.
 - **Integration with other systems:** Receiving signals like vehicle speed from the CAN bus.

2. Wiper Motor

- **Component Name:** Wiper Motor
- **Type:** Actuator
- **Main Function:** This is the primary actuator responsible for physically moving the wiper arms across the windshield. It's typically an electric motor that converts electrical energy into rotational mechanical energy. Its core functions include:
- **Generating Torque:** Providing the force needed to move the wiper arms.
- **Variable Speed Operation:** Often designed to operate at different speeds (low, high) as commanded by the WCM.
- **Park Position Sensing (Internal or External):** Many wiper motors have an integrated limit switch or Hall effect sensor that signals to the WCM when the motor has reached the park position, allowing the WCM to stop power to the motor.

3. Wiper Switch / Stalk Control

-
- **Component Name:** Wiper Switch / Stalk Control
 - **Type:** Input Device / User Interface
 - **Main Function:** This is the primary interface for the driver to control the wiper system. It's typically located on the steering column. Its core functions include:
 - **User Command Input:** Allowing the driver to select different wiper modes (off, intermittent, low speed, high speed, washer).
 - **Signal Generation:** Translating the driver's selections into electrical signals that are sent to the WCM.

4. Wiper Linkage Mechanism

- **Component Name:** Wiper Linkage Mechanism
- **Type:** Mechanical Component (but controlled by actuator)
- **Main Function:** This is a system of rods, pivots, and gears that connect the wiper motor's output shaft to the wiper arms. Its core function is to:
- **Convert Rotary Motion to Linear/Oscillating Motion:** Translate the rotational motion of the wiper motor into the back-and-forth sweeping motion of the wiper arms.
- **Distribute Motion:** Ensure both wiper arms move in sync and cover the desired windshield area.

5. Wiper Arms and Blades

- **Component Name:** Wiper Arms and Blades
- **Type:** Mechanical Component (driven by linkage)
- **Main Function:** These are the visible parts that make contact with the windshield. Their core functions are:
- **Wiper Arms:** Provide the leverage and structure to hold the wiper blades against the windshield.
- **Wiper Blades:** Made of rubber or silicone, they are designed to sweep water and debris away from the windshield.

6. Washer Pump

-
- **Component Name:** Washer Pump
 - **Type:** Actuator
 - **Main Function:** This electric pump is responsible for spraying washer fluid onto the windshield. Its core function is to:
 - **Fluid Delivery:** Pump washer fluid from the reservoir to the washer nozzles.

7. Washer Nozzles

- **Component Name:** Washer Nozzles
- **Type:** Mechanical Component (part of the fluid delivery system)
- **Main Function:** These are small, adjustable openings that direct the spray of washer fluid onto the windshield. Their core function is to:
- **Fluid Dispersion:** Atomize and direct the washer fluid effectively across the windshield.

8. Vehicle Speed Sensor (Indirect Input)

- **Component Name:** Vehicle Speed Sensor (VSS)
- **Type:** Sensor (typically part of the ABS or Powertrain ECU)
- **Main Function:** While not exclusively a wiper system component, the WCM often receives vehicle speed information from the VSS via the vehicle's communication network (e.g., CAN bus). This is crucial for:
- **Intermittent Wipe Speed Adjustment:** The WCM can adjust the frequency of intermittent wipes based on vehicle speed (e.g., slower wipes at higher speeds).

9. Rain Sensor (Optional but common)

- **Component Name:** Rain Sensor
 - **Type:** Sensor
 - **Main Function:** This sensor detects the presence and intensity of rain on the windshield. Its core functions include:
 - **Rain Detection:** Using optical or capacitive principles to sense water droplets on the glass.
-

-
- **Intensity Measurement:** Estimating the amount of rain falling.
 - **Automatic Wiper Activation:** Automatically activating and adjusting the wiper speed based on detected rain.

10. Communication Interfaces (e.g., CAN Bus)

- **Component Name:** Controller Area Network (CAN) Bus
- **Type:** Communication Interface
- **Main Function:** This is the primary network used by most modern automotive ECUs to communicate with each other. For the wiper system, it's essential for:
- **Data Exchange:** The WCM receives data from other ECUs (like vehicle speed from the VSS) and can send status information to other modules.
- **Integration:** Enables the wiper system to be integrated with other vehicle features (e.g., automatic headlights that might also activate wipers in certain conditions).

Summary of Core Elements:

The core elements that define the Wiper System are:

- **Wiper Control Module (WCM/ECU):** The central processing unit.
- **Wiper Motor:** The primary actuator for movement.
- **Wiper Switch/Stalk:** The driver's primary input.
- **Wiper Linkage & Arms/Blades:** The mechanical components that perform the wiping action.
- **Communication Interface (CAN Bus):** For integration and data exchange with other vehicle systems.

Optional but increasingly common core elements include the **Rain Sensor** for automatic operation.

3.4 Item Functions

Reference: 5.4.1 b)

Here are the primary functions of the vehicle-level wiper system, presented concisely:

1. Wiping:

-
- **What it does:** Clears precipitation (rain, snow, etc.) and debris from the windshield.
 - **Conditions:** When precipitation is detected or manually activated.
 - **Inputs:** Rain sensor signal, manual switch position.
 - **Outputs:** Wiper blade movement across the windshield.

2. Speed Control:

- **What it does:** Adjusts the speed of the wiper blades to match the intensity of precipitation.
- **Conditions:** When wiping is active and precipitation levels vary.
- **Inputs:** Rain sensor signal (intensity), manual speed selection.
- **Outputs:** Motor speed command to the wiper motor.

3. Intermittent Wiping:

- **What it does:** Provides periodic wiping at adjustable intervals for light precipitation.
- **Conditions:** When wiping is active and precipitation is light.
- **Inputs:** Rain sensor signal (light precipitation), manual interval selection.
- **Outputs:** Timed activation of the wiper motor.

4. Washer Activation:

- **What it does:** Sprays washer fluid onto the windshield to clean it.
- **Conditions:** When manually activated or automatically triggered by heavy precipitation.
- **Inputs:** Washer switch position, rain sensor signal (heavy precipitation).
- **Outputs:** Activation of the washer pump and nozzle spray.

5. Park Position:

- **What it does:** Returns the wiper blades to their designated resting position at the bottom of the windshield when deactivated.
- **Conditions:** When the wiper system is turned off.
- **Inputs:** Wiper system deactivation signal.
- **Outputs:** Motor command to move blades to the park position.

3.5 Item State Machine

Reference: 5.4.1 b)

Let's define the internal states of a "Wiper System" and their transitions, keeping in mind that these are distinct from the overall vehicle state and are crucial for safety logic.

Wiper System Internal States

These states represent the operational mode and condition of the wiper system itself, independent of whether the vehicle is driving, parked, or in accessory mode.

1. Init (Initialization)

- **Description:** This is the initial state when the wiper system powers up or is reset. It's a transient state where the system performs self-checks, calibrates sensors (if applicable), and prepares for normal operation.

***Characteristics:*

- Wipers are typically in a known, safe position (e.g., parked at the bottom of the windshield).
- No wiper movement is occurring.
- Internal diagnostics are running.
- May involve reading configuration parameters.
- **Safety Relevance:** Ensures the system starts in a predictable and safe state. Prevents unexpected wiper activation during startup.

2. Normal (Operational)

- **Description:** The system is actively responding to user commands or environmental triggers (e.g., rain sensor) and performing its intended function of wiping the windshield.

***Characteristics:*

- Wipers are moving according to the selected speed and pattern (e.g., intermittent, low, high).
 - May be actively monitoring rain sensor input.
 - User interface (stalk controls) is responsive.
 - **Safety Relevance:** This is the primary functional state. Safety logic within this state ensures appropriate wiper speed based on vehicle speed (if applicable), prevents operation in hazardous conditions (e.g., ice buildup), and handles potential sensor failures gracefully.
-

3. Fault (Error)

- **Description:** The wiper system has detected an internal malfunction or an external condition that prevents it from operating safely or as intended.

***Characteristics:*

- Wiper movement may cease abruptly or be disabled.
- An indicator (e.g., dashboard warning light, message) is activated to inform the driver.
- Specific fault codes may be logged.
- The system may attempt a safe fallback mode.
- **Safety Relevance:** This is a critical state for safety. It immediately signals a problem to the driver and prevents potentially dangerous operation (e.g., wipers getting stuck mid-sweep, motor overheating).

4. Sleep (Low Power/Standby)

- **Description:** The wiper system is in a low-power state, conserving energy when not actively needed. It's ready to transition back to Normal operation quickly upon demand.

***Characteristics:*

- Wipers are stationary, typically in the parked position.
- Minimal power consumption.
- System is monitoring for wake-up triggers (e.g., user input, rain sensor activation).
- **Safety Relevance:** While not directly a safety *hazard* state, it's important for overall vehicle system health and battery management. Safety logic ensures it doesn't interfere with critical functions and can wake up reliably when needed.

5. Parked (Safe Position)

- **Description:** The wipers are in their designated resting position, usually at the bottom of the windshield. This is often a sub-state or a specific condition within other states.

***Characteristics:*

- Wipers are stationary.
 - The system is not actively wiping.
 - **Safety Relevance:** Ensures wipers are out of the driver's line of sight and are not obstructing the windshield when not in use. Crucial for preventing distraction and ensuring clear visibility.
-

Transitions Between States

Here's how the Wiper System can transition between these states. These transitions are often triggered by internal logic, external inputs, or system events.

1. Init -> Normal

- **Trigger:** Successful completion of initialization and self-diagnostics.
- **Condition:** No critical faults detected during Init.
- **Safety Logic:** Ensures the system is fully functional and ready before allowing operation.

2. Init -> Fault

- **Trigger:** Detection of a critical fault during initialization (e.g., motor failure, communication error).
- **Condition:** Initialization fails to complete successfully.
- **Safety Logic:** Immediately enters Fault state to prevent operation with a known defect.

3. Normal -> Sleep

- **Trigger:** No user input for a defined period, and no rain sensor activation (if applicable).
- **Condition:** Wipers are in the Parked position.
- **Safety Logic:** Conserves energy when the system is not actively required.

4. Normal -> Fault

***Trigger:*

- Detection of a motor stall or overload.
- Communication loss with the rain sensor.
- Internal component failure detected during operation.
- User command that cannot be executed (e.g., attempting to activate wipers when ice is detected).
- **Condition:** An anomaly is detected during active wiping.
- **Safety Logic:** Immediately stops operation and alerts the driver to a potentially unsafe condition.

5. Normal -> Parked (as a sub-state)

-
- **Trigger:** User deactivates wipers, or rain sensor indicates no rain.
 - **Condition:** Wipers reach the bottom of the windshield.
 - **Safety Logic:** Returns wipers to their safe, out-of-the-way position.

6. Sleep -> Normal

***Trigger:*

- User activates the wiper stalk (e.g., selects a speed).
- Rain sensor detects precipitation.
- **Condition:** System is not in a Fault state.
- **Safety Logic:** Ensures immediate response to driver or environmental needs.

7. Sleep -> Fault

- **Trigger:** A fault is detected *during the wake-up process* (e.g., motor doesn't respond to the initial wake-up command).
- **Condition:** The system fails to transition to Normal operation.
- **Safety Logic:** Prevents operation if a fault is present even in the low-power state.

8. Fault -> Init (Reset)

***Trigger:*

- Ignition cycle (power cycle).
- Specific diagnostic command from a service tool.
- In some systems, a "self-healing" mechanism might attempt a reset after a period.
- **Condition:** The underlying cause of the fault may have been resolved or is being re-evaluated.
- **Safety Logic:** Allows the system to attempt to recover from a fault, but only after a controlled reset. If the fault persists, it will re-enter the Fault state.

9. Fault -> Sleep (Rare/Specific Scenarios)

- **Trigger:** In some very specific fault scenarios, the system might be designed to enter a low-power state to prevent further damage or drain, but this is less common and usually requires a deliberate reset.
- **Condition:** A non-critical fault that allows for power conservation.
- **Safety Logic:** Primarily for system protection rather than immediate driver safety.

10. Parked -> Normal

-
- **Trigger:** User selects a wiper speed or rain sensor detects rain.
 - **Condition:** System is not in Fault or Init.
 - **Safety Logic:** Initiates wiping from the safe position.

Key Considerations for Safety Logic:

- **State Machine Design:** The wiper system's behavior is best modeled as a state machine. Transitions are governed by specific events and conditions.
- **Fault Prioritization:** Critical faults should always lead to the Fault state, disabling operation.
- **Fail-Safe Behavior:** In case of uncertainty or failure, the system should default to a safe state (e.g., wipers off, in the parked position).
- **Driver Notification:** Any transition into or out of the Fault state must be clearly communicated to the driver.
- **Interdependencies:** While distinct from vehicle states, the wiper system's safety logic might consider vehicle speed (e.g., reducing wiper speed at high speeds to prevent damage or noise) or other environmental factors. However, the *internal* states described above are about the wiper system's own condition.
- **Redundancy:** For highly critical applications, there might be redundant sensors or control paths, influencing state transitions.

By clearly defining these internal states and their transitions, engineers can build robust safety logic that ensures the wiper system operates reliably and safely under all conditions.

4. Requirements of the Item

Reference: 5.4.1

4.1 Functional Behaviour at Vehicle Level

Reference: 5.4.1 b)

Wiper System Functional Behavior Across Operating Modes

This document describes the primary functional behavior of the Wiper System during key operating modes, focusing on vehicle-level actions and interactions.

Normal Operating Mode

In this mode, the Wiper System operates as intended, responding to driver commands and environmental conditions to maintain visibility.

FF-01: Manual Wiper Activation & Speed Control

- **Function:** Allows the driver to manually control the operation and speed of the windshield wipers.
- **Driver Interaction:** The driver uses the wiper stalk (typically on the steering column) to select different wiper speeds (e.g., Off, Intermittent, Low, High) and to activate/deactivate the wipers.
- **Conditions:** This function is available when the vehicle's ignition is ON and the wiper system is not in a fault or degraded state.

***Inputs:*

- Driver input from the wiper stalk (position/selection).
- Vehicle speed (for intermittent speed adjustment, if applicable).
- Rain sensor output (if equipped and enabled, to automatically adjust intermittent speed).

***Outputs:*

- Wiper motor activation at the selected speed.
- Indicator on the instrument cluster (e.g., wiper symbol, speed setting).

FF-02: Automatic Wiper Activation (Rain Sensing)

- **Function:** Automatically activates and adjusts the windshield wipers based on detected rainfall.
- **Driver Interaction:** The driver enables or disables the automatic wiper function via a dedicated button or a setting on the wiper stalk. Once enabled, the system operates autonomously.

***Conditions:*

- Vehicle ignition is ON.
 - Automatic wiper function is enabled by the driver.
 - Rain sensor detects sufficient moisture on the windshield.
 - Wiper system is not in a fault or degraded state.
-

***Inputs:*

- Rain sensor output (detecting water droplets and intensity).
- Vehicle speed (to adjust intermittent wipe frequency).
- Driver selection to enable/disable automatic mode.

***Outputs:*

- Wiper motor activation at an automatically determined speed (intermittent, low, or high, depending on rain intensity and vehicle speed).
- Indicator on the instrument cluster (e.g., "Auto" mode engaged).

FF-03: Wiper Park Position

- **Function:** Ensures that the windshield wipers return to their designated "parked" position at the bottom of the windshield when deactivated.
- **Driver Interaction:** This function is automatic and requires no direct driver interaction. The driver simply turns the wipers off.

***Conditions:*

- Wiper system is activated and then deactivated by the driver.
- Wiper motor has completed its current sweep.
- Wiper system is not in a fault or degraded state.

***Inputs:*

- Wiper motor position feedback (to detect the parked position).
- Driver deactivation command.

***Outputs:*

- Wiper motor continues to run until the wipers reach the park position, then stops.

Fault Operating Mode

In this mode, the Wiper System experiences a malfunction that prevents it from operating correctly or safely. The system will typically default to a safe state or provide clear indications to the driver.

FF-04: Wiper System Fault Indication

- **Function:** Informs the driver that a fault has been detected within the wiper system.
- **Driver Interaction:** The driver observes visual and/or audible warnings. They may attempt to operate the wipers, but the system will not function normally.

***Conditions:*

- Detection of a critical fault within the wiper control module, motor, or associated sensors.
- Examples: Wiper motor failure, communication loss with the control module, sensor malfunction preventing safe operation.

***Inputs:*

- Internal diagnostic signals from the wiper control module.
- Communication bus status.

***Outputs:*

- Illumination of a dedicated Wiper Fault warning light on the instrument cluster.
- Potentially an audible chime.
- Wiper system is disabled or operates in a severely limited capacity.

FF-05: Wiper System Failure to Operate

- **Function:** The windshield wipers are completely inoperable due to a fault.
- **Driver Interaction:** The driver attempts to activate the wipers, but they do not move. The driver will rely on other visibility aids (e.g., defroster, manual cleaning) and may need to pull over if visibility is compromised.

***Conditions:*

- A fault condition (as described in FF-04) prevents any wiper operation.
- This is the direct consequence of a fault.

***Inputs:*

- Driver input from the wiper stalk (attempting activation).
- Internal fault detection signals.

***Outputs:*

-
- No wiper motor activation.
 - Wiper Fault warning light (as per FF-04).

Degraded Operating Mode

In this mode, the Wiper System experiences a non-critical fault or a limitation that allows for some functionality, but not at its full intended performance. The system will often operate at a reduced capability or with specific limitations.

FF-06: Reduced Wiper Speed/Functionality

- **Function:** The wiper system operates at a reduced speed or with limited functionality due to a non-critical fault or a specific operational constraint.
- **Driver Interaction:** The driver may notice that the wipers are slower than usual, or that certain speed settings are unavailable. They may still be able to activate the wipers, but with reduced effectiveness.

***Conditions:*

- Partial failure of the wiper motor speed control.
- Intermittent communication issues with the wiper control module.
- A sensor providing degraded but usable data (e.g., a partially functioning rain sensor).
- Vehicle operating in a specific mode that limits wiper function (e.g., very low speed operation in certain parking maneuvers).

***Inputs:*

- Degraded sensor data.
- Partial failure signals from the wiper control module.
- Driver input from the wiper stalk.

***Outputs:*

- Wiper motor activation at a slower-than-normal speed.
 - Only certain speed settings (e.g., Low, but not High) may be available.
 - Automatic wiper function may be disabled or operate erratically.
 - Potentially a "Wiper System Limited" indicator on the instrument cluster (less severe than a full fault).
-

FF-07: Intermittent Wiper Functionality with Limited Adjustment

- **Function:** The intermittent wiper function is available, but the driver has limited or no ability to adjust the wipe interval, or the automatic adjustment based on rain intensity is disabled.
- **Driver Interaction:** The driver can activate the intermittent mode, but the speed of the wipes may be fixed or not respond to changes in rain. If automatic rain sensing is part of the degraded mode, it may be disabled or operate at a default interval.

***Conditions:*

- Fault in the intermittent speed control circuit.
- Rain sensor malfunction that prevents automatic adjustment but allows basic intermittent operation.
- Communication issue with the rain sensor.

***Inputs:*

- Driver input to select intermittent mode.
- Degraded rain sensor data or lack of communication.
- Wiper control module operating with limited parameters.

***Outputs:*

- Wipers operate in an intermittent pattern.
- Wipe interval may be fixed or not adjustable by the driver.
- Automatic speed adjustment based on rain intensity is disabled.

4.2 Required Quality, Performance, and Availability

Reference: 5.4.1 c)

Here are critical quantitative performance requirements for a Wiper System, focusing on safety-relevant metrics, along with their safety rationale:

Wiper System: Critical Quantitative Performance Requirements (Safety-Relevant)

1. Accuracy

Definition: The ability of the wiper blades to effectively clear the windshield of precipitation (rain, snow, etc.) and debris to provide a clear field of vision for the driver.

Quantitative Requirement:

- **Rain Clearing Efficiency:** The wiper system shall clear at least **95%** of the windshield surface area within **3 full sweeps** at its highest operational speed setting when subjected to a simulated rainfall rate of **50 mm/hour**.
- **Debris Removal:** The wiper system shall remove at least **90%** of simulated dry debris (e.g., leaves, small twigs) from the primary driver's field of vision within **5 full sweeps** at its highest operational speed setting.

Safety Rationale:

- **Reduced Visibility:** Inadequate clearing of the windshield directly impairs the driver's ability to see the road, other vehicles, pedestrians, and potential hazards. This is a primary cause of accidents.
- **Sudden Obstructions:** Heavy rain or accumulated debris can create sudden and significant reductions in visibility, requiring immediate and effective clearing.
- **Maintaining Control:** A clear view is essential for maintaining vehicle control, especially during critical maneuvers like braking, steering, and lane changes.
- **Night Driving & Glare:** Ineffective wiping can lead to water streaks or debris that, when illuminated by headlights or sunlight, can cause glare and further reduce visibility, particularly at night or in adverse weather.

2. Response Time

Definition: The time taken for the wiper system to initiate movement and reach its operational speed after activation by the driver or an automatic sensor.

Quantitative Requirement:

- **Activation Latency:** The wiper blades shall begin to move within **0.5 seconds** of the wiper control being activated (manual or automatic).
- **Speed Attainment:** The wiper blades shall reach their designated operational speed (e.g., high speed) within **1.5 seconds** of initiating movement.

Safety Rationale:

-
- **Immediate Hazard Response:** In situations of sudden downpours or unexpected debris, the driver needs immediate assistance in clearing the windshield. Delays can mean the difference between seeing a hazard and not.
 - **Preventing "Blind" Moments:** A rapid response prevents a period of reduced visibility during which the driver is effectively "blind" to their surroundings.
 - **Automatic Systems:** For automatic wiper systems that react to rain, a quick response is crucial to prevent the windshield from becoming obscured before the system engages.

3. Availability

Definition: The proportion of time the wiper system is operational and ready for use when needed. This includes the system's ability to be activated and function as intended.

Quantitative Requirement:

- **Operational Availability:** The wiper system shall have an operational availability of at least 99.9% over its expected service life. This means the system should be functional and capable of being activated when the driver attempts to use it.
- **Component Reliability (Implicit):** While not a direct availability metric, the reliability of individual components (motor, switch, linkage) directly contributes to overall availability.

Safety Rationale:

- **Consistent Functionality:** The driver must be able to rely on the wiper system to function whenever weather conditions demand it. Unavailability during adverse weather is a critical safety failure.
- **Preventing Unexpected Failures:** A high availability requirement implies robust design and manufacturing to minimize unexpected failures that could leave the driver without this essential safety feature.
- **Emergency Situations:** In emergency braking or evasive maneuvers, the driver needs a clear windshield. The system's availability ensures it's ready for such critical moments.

4. Reliability

Definition: The probability that the wiper system will perform its intended function without failure for a specified period of time or number of operational cycles.

Quantitative Requirement:

-
- **Mean Time Between Failures (MTBF):** The wiper system shall have an MTBF of at least **10,000 operational hours** or **500,000 wiper cycles**, whichever is more stringent, under typical operating conditions.
 - **Failure Rate:** The failure rate of the wiper system shall not exceed **1 x 10⁻⁵ failures per operational hour**.

Safety Rationale:

- **Preventing Catastrophic Failures:** A highly reliable system is less likely to fail unexpectedly, which could lead to a sudden loss of visibility during critical driving situations.
- **Long-Term Performance:** Drivers rely on their vehicles for extended periods. High reliability ensures the wiper system remains functional throughout the vehicle's lifespan, not just for a short initial period.
- **Reduced Maintenance & Unforeseen Issues:** Reliable systems require less frequent maintenance and are less prone to unexpected breakdowns that could compromise safety.
- **Component Durability:** Reliability requirements drive the selection of durable components that can withstand the stresses of operation in various environmental conditions (temperature, moisture, vibration).

Important Considerations for Wiper System Performance:

- **Environmental Conditions:** These requirements should be considered under a range of environmental conditions, including extreme temperatures, humidity, and exposure to road salt and debris.
- **Driver Interaction:** The ease of use and intuitiveness of the wiper controls are also safety-relevant, though harder to quantify directly.
- **System Integration:** The performance of the wiper system is also dependent on its integration with other vehicle systems, such as the rain sensor and the vehicle's electrical system.

By defining and meeting these quantitative performance requirements, manufacturers can ensure that the wiper system effectively contributes to driver safety by maintaining a clear and unobstructed view of the road.

4.3 Constraints Regarding the Item

Reference: 5.4.1 d)

Here's a breakdown of the critical constraints for a Wiper System, categorized as requested:

Wiper System Critical Constraints

Environmental Constraints

| Constraint | Operating Range/Consequence |

|---|---|

| **Temperature** | **Range:** Typically -40°C to +85°C (automotive standard).
 Consequence: Extreme temperatures can lead to:
 Low Temp: Lubricant thickening, motor sluggishness, material embrittlement (plastic parts), ice formation affecting movement.
 High Temp: Overheating of motor and electronics, reduced lifespan of components, potential for thermal runaway, material degradation. |

| **Vibration** | **Range:** Varies significantly based on vehicle type and location (e.g., engine bay vs. cabin). Can be sinusoidal or random, with specific G-force limits.
 Consequence: Constant vibration can cause:
 Mechanical Fatigue: Cracking or failure of plastic components, loosening of fasteners, wear on moving parts.
 Electrical Issues: Intermittent connections, damage to solder joints, sensor malfunction.
 *Noise:** Increased operational noise due to rattling or loose parts. |

| **EMC (Electromagnetic Compatibility)** | **Range:** Must meet automotive EMC standards (e.g., CISPR 25, ISO 11452).
 Consequence:
 Emissions: The wiper system itself can generate electromagnetic interference (EMI) that disrupts other vehicle electronics (radio, navigation, control modules).
 Immunity: The wiper system can be susceptible to external EMI, leading to erratic operation, false sensor readings, or complete failure. |

| **IP (Ingress Protection)** | **Range:** Typically IP54 or higher for exposed components (e.g., motor housing, linkage).
 Consequence:
 Water/Dust Ingress: Can lead to corrosion of electrical contacts, motor damage, bearing failure, and short circuits.
 Salt/Chemical Exposure: In regions with road salt or harsh cleaning chemicals, higher IP ratings are crucial to prevent material degradation and corrosion. |

Functional Dependencies

| Dependency | Operating Range/Consequence |

|---|---|

| **Sensors** | **Range:** Dependent on sensor type (e.g., rain sensor, speed sensor, park position sensor).
 Consequence:
 Rain Sensor: Inaccurate or no detection of rain leads to wipers not activating or operating at the wrong speed.
 Speed Sensor: Incorrect wiper speed or failure to adjust speed based on vehicle speed.
 Park Position Sensor: Wipers not returning to their resting position, potentially obstructing view or causing damage.
 Obstruction Sensor: Failure to detect an obstruction can lead to motor damage or wiper arm breakage. |

| **Power** | **Range:** Typically 12V DC automotive electrical system. Voltage fluctuations are common.
 Consequence:
 Low Voltage: Motor may not have enough torque to operate, or operation may be slow and erratic.
 High Voltage: Can damage motor windings and electronic control units.
 *Intermittent Power:** Wipers may stop unexpectedly or operate erratically. |

| **Bus (e.g., CAN Bus)** | **Range:** Standard automotive communication protocols.
 Consequence:
 Communication Errors: If the wiper system relies on a bus for commands or sensor data, communication failures can lead to:
 No operation: If commands are not received.
 Erratic operation: If incorrect data is received.
 Inability to diagnose: If diagnostic messages cannot be transmitted or received. |

Resource Limits

| Limit | Operating Range/Consequence |

|---|---|

| **Power Consumption** | **Range:** Varies based on motor size, speed, and load (e.g., heavy rain, ice). Typically a few amps.
 Consequence:
 Excessive Consumption: Can drain the vehicle battery, especially during prolonged use or when other high-power accessories are active.
 Under-spec'd Power: May lead to insufficient torque for effective wiping, especially under load. |

| **Space** | **Range:** Limited by vehicle design, often in the cowl area or behind the dashboard.
 Consequence:
 Component Size: Restricts the size and power of the motor, gearbox, and control electronics.
 Packaging Complexity: Requires careful design to ensure all components fit and are accessible for maintenance.
 *Aerodynamics:** Wiper arms and blades can create drag and noise if not designed for optimal airflow. |

| **CPU/Processing Power** | **Range:** Typically a low-power microcontroller for basic wiper control, or a more powerful ECU for advanced features (e.g., rain sensing, variable speed).
 Consequence:
 Insufficient Processing: For advanced features like complex rain sensing

algorithms or adaptive wiping, a low-power CPU may lead to slow response times, inaccurate operation, or inability to implement desired functionality.
 Overhead: Even for basic functions, the CPU needs to manage motor control, sensor inputs, and communication, requiring a minimum level of processing capability. |

4.4 Potential Consequences of Behavioural Shortfalls

Reference: 5.4.1 e)

Here's a breakdown of potential hazardous events for a Wiper System, formatted for HARA, based on the assumption of typical critical functions:

Critical Function: Wiper Blade Movement (Wiping Action)

- **[Malfunction]** Wiper blades fail to move or move erratically.
- **[Hazardous Event]** Obscured driver's vision due to precipitation (rain, snow, sleet).
- **[Potential Harm]** Loss of vehicle control, collision with other vehicles, pedestrians, or fixed objects.
- **[Malfunction]** Wiper blades move at an inappropriate speed (too slow or too fast for conditions).
- **[Hazardous Event]** Inadequate clearing of precipitation, leading to partially obscured driver's vision.
- **[Potential Harm]** Delayed reaction time, increased risk of collision.
- **[Malfunction]** Wiper blades stop in the driver's line of sight.
- **[Hazardous Event]** Complete obstruction of driver's vision.
- **[Potential Harm]** Immediate loss of vehicle control, high risk of severe collision.

Critical Function: Wiper Park Position

- **[Malfunction]** Wiper blades fail to return to their designated park position.
- **[Hazardous Event]** Wiper blades remain in the driver's line of sight when not actively wiping.
- **[Potential Harm]** Obscured driver's vision, distraction, potential for collision.

Critical Function: Wiper Speed Control (Intermittent, Low, High)

- **[Malfunction]** Wiper system defaults to a single, inappropriate speed (e.g., always high speed in light drizzle, or always slow in heavy downpour).
 - **[Hazardous Event]** Inadequate clearing of precipitation or excessive noise/distraction.
 - **[Potential Harm]** Reduced visibility leading to collision, driver distraction.
-

-
- **[Malfunction]** Wiper system fails to adjust speed based on sensor input (if applicable, e.g., rain sensor).
 - **[Hazardous Event]** Inappropriate wiping speed for current precipitation levels.
 - **[Potential Harm]** Reduced visibility leading to collision, driver distraction.

Critical Function: Washer Fluid Dispensing

- **[Malfunction]** Washer fluid pump fails to operate.
- **[Hazardous Event]** Inability to clear dirt, mud, or debris from the windshield, especially in conjunction with precipitation.
- **[Potential Harm]** Severely obscured driver's vision, leading to loss of control and collision.
- **[Malfunction]** Washer fluid is dispensed continuously or at inappropriate times.
- **[Hazardous Event]** Distraction to the driver, potential for temporary windshield obscuration if fluid is applied during critical driving maneuvers.
- **[Potential Harm]** Driver distraction, momentary loss of visibility leading to increased risk of collision.

Critical Function: Wiper Motor Operation

- **[Malfunction]** Wiper motor seizes or fails to provide sufficient torque.
- **[Hazardous Event]** Wiper blades stop moving or move very slowly, unable to effectively clear precipitation.
- **[Potential Harm]** Obscured driver's vision, leading to loss of control and collision.
- **[Malfunction]** Wiper motor overheats and fails.
- **[Hazardous Event]** Sudden cessation of wiper blade movement during precipitation.
- **[Potential Harm]** Obscured driver's vision, leading to loss of control and collision.

Critical Function: Wiper Control Switch/Actuator

- **[Malfunction]** Wiper control switch fails to respond to driver input.
 - **[Hazardous Event]** Inability to activate or deactivate wipers, or change their speed, as needed for visibility.
 - **[Potential Harm]** Obscured driver's vision, leading to loss of control and collision.
 - **[Malfunction]** Wiper control switch activates wipers unintentionally.
 - **[Hazardous Event]** Unwanted wiper movement, potentially distracting the driver or obscuring vision if activated at the wrong time.
 - **[Potential Harm]** Driver distraction, momentary loss of visibility leading to increased risk of collision.
-

Critical Function: Wiper Linkage System

- **[Malfunction]** Wiper linkage breaks or becomes disconnected.
- **[Hazardous Event]** Wiper blades become inoperative or move erratically.
- **[Potential Harm]** Obscured driver's vision, leading to loss of control and collision.

Note: This list assumes a standard wiper system. More advanced systems with features like rain sensors, heated wipers, or integrated washer nozzles would have additional critical functions and corresponding hazardous events. The "Functional Behaviour at Vehicle Level" section would need to be consulted for a comprehensive list.

4.5 Capabilities of the Actuators

Reference: 5.4.1 f)

To quantify critical actuation capabilities of a "Wiper System" relevant for a Hazard Analysis and Risk Assessment (HARA), we need to consider the functions and potential failure modes that could lead to hazards. The specific numbers will depend heavily on the application (e.g., automotive, aerospace, industrial machinery), but I will provide **representative examples and ranges** that are commonly considered.

It's crucial to understand that a "Wiper System" is a broad term. For a HARA, you'd need to define the specific components and their intended functions. For this example, let's assume a typical automotive windshield wiper system.

Here's a breakdown of critical actuation capabilities and their quantification, along with specific numbers where applicable:

Critical Actuation Capabilities of a Wiper System for HARA

The primary actuation capabilities of a wiper system relate to its ability to move the wiper blades across the windshield at the correct speed and with sufficient force to clear obstructions.

1. Wiper Motor Actuation (Primary Movement)

This is the core actuation capability. It involves the motor's ability to generate torque and rotate the wiper linkage.

- **Function:** To move wiper blades across the windshield to clear precipitation and debris.

***Critical Parameters:*

- **Speed/Frequency:** The rate at which the wiper blades sweep the windshield. This is crucial for visibility.

***Quantification:*

- **Minimum Sweep Frequency (for effective clearing):** Typically around **20-30 sweeps per minute (SPM)**. Below this, visibility can be compromised in moderate rain.
- **Maximum Sweep Frequency (for comfort/noise):** Often around **60-80 SPM**. Higher speeds can be noisy and unnecessary.
- **Response Time (Start-up):** The time it takes for the motor to reach its operational speed from a standstill. This is important for rapid clearing when rain suddenly intensifies.
- **Specific Number Example: Contactor/Relay Activation to Full Speed: < 100 ms** (This is a general electronic control system response time. The mechanical inertia of the motor and linkage will add to this).
- **Specific Number Example: Full Sweep Cycle Initiation to Blade Movement: < 50 ms** (This refers to the mechanical response once the electrical signal is received).
- **Torque Output:** The rotational force the motor can generate. This is critical for overcoming friction, ice, and snow.

***Quantification:*

- **Minimum Operating Torque (under load):** This is highly dependent on wiper size, blade design, and environmental conditions. For automotive, it might be in the range of **1-5 Nm** at the motor shaft.
- **Stall Torque (maximum allowable):** The torque the motor can produce before stalling or overheating. This is a safety limit.
- **Specific Number Example: Stall Torque Limit: > 10 Nm** (This is a design parameter to prevent damage to the motor or linkage if the wipers are obstructed).
- **Directional Control:** The ability to reverse the wiper blade motion at the ends of the sweep.

***Quantification:*

- **Reversal Time:** The time taken to reverse direction.
 - **Specific Number Example: Time to Reverse Direction: < 200 ms** (This includes motor braking and acceleration in the opposite direction).
-

2. Wiper Linkage Actuation (Mechanical Transmission)

The linkage translates the motor's rotational motion into the oscillating motion of the wiper arms.

- **Function:** To transmit and convert the motor's torque to the wiper arms.

***Critical Parameters:*

- **Mechanical Strength/Integrity:** The linkage must withstand the forces applied by the motor and external loads.

***Quantification:*

- **Maximum Load Capacity (without permanent deformation):** This is a structural integrity parameter.
- **Specific Number Example: Linkage Yield Strength:** Designed to withstand forces significantly higher than expected operating loads, e.g., > 50 Nm applied to the linkage arms.
- **Force Limits (for safety/protection):** The linkage might have designed weak points or slip clutches to prevent damage to the motor or windshield in case of severe obstruction.
- **Specific Number Example: Slip Clutch Activation Force: 3-7 Nm** (This is the torque at the wiper arm pivot that will cause the clutch to slip, protecting the motor).
- **Backlash/Play:** Excessive play can lead to jerky motion and reduced clearing effectiveness.

***Quantification:*

- **Maximum Allowable Backlash:** Typically measured in degrees or millimeters at the wiper arm.
- **Specific Number Example: Maximum Backlash at Wiper Arm Pivot: < 2 degrees or < 0.5 mm.**

3. Wiper Motor Control System Actuation (Electrical/Electronic)

This encompasses the control signals and power delivery to the motor.

- **Function:** To command the motor to operate at specific speeds, directions, and to respond to user inputs or sensor data.
-

***Critical Parameters:*

- **Response Time (Signal to Actuator):** The time from a command being issued (e.g., user presses button, rain sensor detects water) to the motor receiving power and starting to move.

***Quantification:*

***User Input to Motor Activation:*

- **Specific Number Example: Button Press to Wiper Movement Initiation: < 150 ms.**

***Sensor Input to Motor Activation:*

- **Specific Number Example: Rain Sensor Detection to Wiper Movement Initiation: < 200 ms** (allowing for sensor processing and decision making).
- **Power Delivery Reliability:** Consistent and sufficient power to the motor.

***Quantification:*

***Voltage Drop Limits:*

- **Specific Number Example: Maximum Allowable Voltage Drop at Motor Terminals under Load: < 5%** of nominal voltage.
- **Contact/Relay Response Time:** If a contactor or relay is used to switch motor power.

***Quantification:*

- **Specific Number Example: Contactor Opens/Closes in: < 30 ms** (This is a typical specification for automotive-grade relays).

4. Wiper Blade Actuation (Contact with Surface)

While not strictly "actuation" in the motor sense, the blade's ability to press against the windshield is a critical outcome of the actuation system.

- **Function:** To make consistent contact with the windshield to wipe away water.

***Critical Parameters:*

- **Contact Force (Pressure):** The force exerted by the wiper blade on the windshield.
-

***Quantification:*

***Minimum Contact Force (for effective wiping):*

- **Specific Number Example: Minimum Blade Contact Force: 0.5 - 1.5 N per cm of blade length.**

***Maximum Contact Force (to prevent damage):*

- **Specific Number Example: Maximum Blade Contact Force: < 3 N per cm of blade length (to avoid scratching the glass).**

HARA Relevance and Examples of Hazards

These quantified capabilities are critical for HARA because their failure or degradation can lead to hazards. Here are some examples:

***Hazard:** **Reduced Visibility due to Ineffective Wiping.*

- **Cause:** Wiper motor fails to reach sufficient speed (below 20 SPM), or contactor fails to close, or linkage is too weak and stalls under load.
- **HARA Relevance:** Quantified parameters like **Minimum Sweep Frequency** and **Stall Torque Limit** are directly relevant.

***Hazard:** **Sudden Loss of Visibility due to Wiper System Failure.*

- **Cause:** Wiper motor suddenly stops or linkage breaks.
- **HARA Relevance:** Quantified parameters like **Stall Torque Limit** and **Linkage Yield Strength** are relevant for understanding failure modes.

***Hazard:** **Driver Distraction or Startle Response due to Jerky or Unpredictable Wiper Motion.*

- **Cause:** Excessive backlash in the linkage, or erratic motor control.
- **HARA Relevance:** Quantified parameters like **Maximum Allowable Backlash** are relevant.

***Hazard:** **Damage to Windshield or Wiper System Components.*

- **Cause:** Wiper motor attempts to overcome an obstruction beyond its stall torque, or linkage is not designed with protective features.
 - **HARA Relevance:** Quantified parameters like **Stall Torque Limit** and **Slip Clutch Activation Force** are crucial.
-

***Hazard:** **Delayed Response to Sudden Rain.*

- **Cause:** Slow response time of the control system or motor to initiate wiping.
- **HARA Relevance:** Quantified parameters like **User Input to Motor Activation** and **Sensor Input to Wiper Movement Initiation** are critical.

In summary, when performing a HARA for a wiper system, you would:

1. **Identify the critical functions** (e.g., clearing windshield, responding to commands).
2. **Determine the key actuation parameters** for each function (speed, force, response time).
3. **Quantify these parameters with specific, measurable numbers**, often derived from design specifications, testing, or industry standards.
4. **Analyze potential failure modes** of these actuations and their impact on safety.

The numbers provided above are illustrative. A real HARA would require detailed engineering data specific to the wiper system being analyzed.

5. Boundary of the Item, Interfaces, and Interaction Assumptions

Reference: 5.4.2

5.1 Elements of the Item

Reference: 5.4.2 a)

Wiper System Boundary Definition

The **Wiper System** is defined as the collection of all components directly responsible for the operation, control, and monitoring of the vehicle's windshield wipers. Its primary function is to clear the windshield of precipitation and debris to ensure driver visibility.

In-Scope Components:

The following components are considered part of the Wiper System:

1. Electronic Control Unit (ECU):

- **Wiper Control Module (WCM) / Wiper ECU:** This is the central processing unit of the wiper system. It receives inputs from sensors and user commands, processes this information, and sends commands to actuators. It manages wiper speed, park position, intermittent operation, and potentially other features like rain sensing.

2. Communication Interfaces:

- **Controller Area Network (CAN) Bus Interface:** The WCM communicates with other ECUs in the vehicle (e.g., Body Control Module (BCM), Powertrain Control Module (PCM), Body Domain Controller (BDC)) via the CAN bus. This allows for receiving vehicle speed information, ignition status, and potentially diagnostic data.
- **Local Interconnect Network (LIN) Bus Interface (if applicable):** Some simpler wiper systems might use a LIN bus for communication with specific components like the wiper motor itself or a dedicated rain sensor.

3. Sensors:

- **Rain Sensor:** Detects the presence and intensity of rain on the windshield. This is a key input for automatic wiper operation.
- **Wiper Motor Position Sensor (e.g., Hall effect sensor, encoder):** Provides feedback to the WCM on the current position of the wiper blades, particularly for ensuring they return to the park position.
- **Vehicle Speed Sensor (Input via CAN bus):** While not a direct component of the wiper system itself, the vehicle speed is a crucial input for adjusting wiper speed (e.g., faster wipers at higher speeds). This is considered an *interface* to an external system.
- **Ignition Status Sensor (Input via CAN bus):** Indicates whether the vehicle's ignition is on or off, controlling the overall power supply to the wiper system. This is considered an *interface* to an external system.

4. Actuators:

- **Wiper Motor:** The electric motor that drives the wiper arms and blades.
- **Wiper Relay(s):** Switches that control the power supply to the wiper motor, enabling different speeds and operation modes.
- **Washer Pump Motor:** The electric motor that pumps washer fluid to the windshield.
- **Washer Pump Relay(s):** Switches that control the power supply to the washer pump motor.

5. User Interface Components (directly interacting with the driver):

- **Wiper Stalk/Lever:** The physical control on the steering column used by the driver to select wiper speed, intermittent settings, and activate the washer function.
- **Indicator Lights/LEDs (if integrated into the wiper control):** For displaying wiper status or settings.

Explicitly EXCLUDED Components:

The following components are **NOT** considered part of the Wiper System for the purpose of defining its safety responsibility:

- **Battery Cells:** The power source for the entire vehicle. The Wiper System relies on the vehicle's electrical system, but the battery cells themselves are managed by the Battery Management System (BMS) or the vehicle's overall electrical architecture.
- **Alternator/Generator:** Responsible for charging the battery and supplying power to the vehicle's electrical systems.
- **Vehicle Wiring Harness (beyond the direct connections to Wiper System components):** While the wiring harness is essential for connectivity, the broader harness infrastructure is part of the vehicle's overall electrical architecture, not solely the Wiper System.
- **Fuse Box/Circuit Breakers (unless specifically dedicated to the Wiper System and integrated within its control logic):** These are general protection devices for the vehicle's electrical system.
- **External Cloud Services:** Any remote diagnostics, over-the-air updates, or data logging services that are not directly processed by the WCM.
- **Windshield Washer Fluid Reservoir and Tubing:** These are mechanical components for fluid delivery, not part of the electronic or electromechanical control system.
- **Wiper Blades and Arms:** These are the physical components that perform the wiping action. While essential for the *function* of wiping, they are considered mechanical end-effectors and not part of the *control system* itself.
- **Other ECUs (e.g., BCM, PCM, BDC, ADAS ECU):** While these ECUs may provide inputs to or receive outputs from the Wiper System (e.g., vehicle speed, rain intensity from an ADAS camera), they are distinct systems with their own boundaries and safety responsibilities.
- **Diagnostic Tools and Testers:** External equipment used for troubleshooting and maintenance.
- **Vehicle Chassis and Body Structure:** The physical framework of the vehicle.
- **Driver's Eyesight and Judgment:** The ultimate responsibility for safe driving rests with the driver.

Rationale for Boundary Definition and Safety Responsibility:

This clear boundary definition is crucial for assigning safety responsibility. The **Wiper System** is responsible for the correct and reliable functioning of its in-scope components to ensure the driver has adequate visibility under various weather conditions. Any failure within the defined Wiper System components that leads to a loss of visibility and a subsequent safety incident

would fall under the safety responsibility of the Wiper System's design, manufacturing, and maintenance.

By explicitly excluding external systems and components, we avoid ambiguity and ensure that the safety analysis and responsibility are focused on the specific elements that directly control and execute the wiper function. This allows for targeted risk assessments, validation, and mitigation strategies for the Wiper System itself.

5.2 Assumptions Concerning Effects on the Vehicle

Reference: 5.4.2 b)

When performing a Hazard Analysis and Risk Assessment (HARA) for a vehicle's wiper system, it's crucial to make explicit assumptions about how the driver will perceive and react to the system's outputs. These assumptions directly influence the severity and controllability of potential hazards. Here are critical assumptions about vehicle/driver response to 'Wiper System' outputs that support HARA scenarios:

I. Driver Perception and Awareness Assumptions:

***Auditory Warnings:*

- **Audibility:** The driver will hear an auditory warning (e.g., chime, beep) if it is within a reasonable volume range for the ambient noise level of the vehicle (e.g., typical road noise, HVAC operation).
- **Distinctiveness:** The auditory warning will be distinct enough from other vehicle sounds and external noises to be recognized as a specific system alert.
- **Comprehensibility:** The driver understands the meaning of the auditory warning (e.g., "wiper system fault," "wiper fluid low"). This often relies on prior training, user manuals, or standardized warning sounds.
- **Timeliness of Perception:** The driver will perceive the auditory warning within a specified timeframe after it is activated (e.g., within 1-2 seconds).

***Visual Warnings (Instrument Cluster/Display):*

- **Visibility:** The driver will see a visual warning (e.g., icon, text message) on the instrument cluster or infotainment display if their line of sight is directed towards it.
- **Legibility:** The visual warning will be legible under various lighting conditions (day/night, direct sunlight) and at typical viewing distances.

-
- **Distinctiveness:** The visual warning will be distinct from other information displayed and easily identifiable as a system alert.
 - **Comprehensibility:** The driver understands the meaning of the visual warning (e.g., "wiper system malfunction," "wiper blades worn").
 - **Timeliness of Perception:** The driver will perceive the visual warning within a specified timeframe after it is activated (e.g., within 2-3 seconds).

***Haptic Feedback (if applicable):*

- **Perceptibility:** The driver will feel haptic feedback (e.g., vibration) if it is applied to a control they are interacting with or a relevant area.
- **Distinctiveness:** The haptic feedback will be distinct enough to be recognized as a system-related event.
- **Timeliness of Perception:** The driver will perceive the haptic feedback within a specified timeframe after it is activated (e.g., within 0.5-1 second).

II. Driver Reaction and Action Assumptions:

***Reaction Time:*

- **Standard Reaction Time:** The driver will initiate a physical or mental response to a perceived warning within a typical human reaction time (e.g., 0.5 to 2 seconds, depending on the complexity of the warning and the driver's current state).
- **Distraction Impact:** Reaction times may be longer if the driver is significantly distracted (e.g., engaged in conversation, looking at navigation). HARA scenarios should consider these variations.

***Understanding of System Functionality:*

- **Basic Operation:** The driver understands the basic function of the wiper system (e.g., how to turn them on/off, adjust speed, activate washers).
- **Implications of Failure:** The driver understands the potential consequences of a wiper system failure (e.g., reduced visibility in rain, potential for damage to wipers).

***Driver's Current State:*

- **Alertness:** The driver is generally alert and attentive to the driving task. HARA should consider scenarios where the driver is fatigued, drowsy, or under the influence of substances.
-

-
- **Cognitive Load:** The driver's cognitive load is within a manageable range. High cognitive load (e.g., complex navigation, stressful traffic) can impair reaction times and decision-making.
 - **Physical State:** The driver's physical capabilities are unimpaired (e.g., no hand injuries preventing them from operating controls).

***Driver's Intent and Prioritization:*

- **Safety Priority:** The driver prioritizes safety and will take action to mitigate hazards that compromise visibility.
- **Actionable Response:** The driver will attempt to take an appropriate action based on the warning and their understanding of the system. This could include:
- **Troubleshooting:** Attempting to reset the system, check fuses, or consult the owner's manual.
- **Seeking Assistance:** Pulling over to a safe location, contacting roadside assistance.
- **Adapting Driving:** Slowing down, increasing following distance, or stopping if visibility is severely compromised.

***Interaction with Other Systems:*

- **No Conflicting Information:** The wiper system's outputs do not conflict with or confuse the driver's understanding of other vehicle systems.
- **Synergistic Warnings:** If multiple warnings occur simultaneously, the driver can still prioritize and understand them.

III. Assumptions about System Behavior and Failure Modes:

- **Predictable Failure Modes:** The wiper system's failure modes are understood and can be categorized (e.g., complete failure, intermittent operation, stuck in position).
- **Clear Indication of Failure:** When a failure occurs, the system provides a clear and unambiguous indication to the driver.
- **No Unintended Activation:** The wiper system does not activate unexpectedly or in a manner that could surprise or distract the driver.

Examples of how these assumptions support HARA scenarios:

***Scenario: Wiper system fails completely during heavy rain.*

- **Assumption:** "Driver will perceive a visual warning on the instrument cluster within 3 seconds."

-
- **Assumption:** "Driver understands that complete wiper failure in heavy rain leads to severely reduced visibility."
 - **Assumption:** "Driver will react to the warning and attempt to pull over to a safe location within 10 seconds."
 - **HARA Impact:** This assumption helps determine the controllability of the hazard. If the driver reacts quickly and appropriately, the severity of the hazard (e.g., collision due to poor visibility) can be reduced.

***Scenario: Wiper fluid low warning is ignored.*

- **Assumption:** "Driver is currently engaged in a complex conversation and does not immediately perceive the auditory warning."
- **Assumption:** "Driver prioritizes completing the conversation over immediate system checks."
- **HARA Impact:** This assumption highlights a potential controllability issue. If the driver's priorities or distractions prevent them from reacting, the hazard (e.g., windshield becoming obscured by dirt and debris) becomes more severe.

By clearly defining these assumptions, HARA practitioners can build more robust and realistic safety cases for wiper systems, ensuring that potential hazards are adequately identified, analyzed, and mitigated. It's important to note that these are *assumptions* and should be validated through testing, user studies, and consideration of real-world driving conditions.

5.3 Functionality Required by Other Items

Reference: 5.4.2 c)

The Wiper System, while seemingly simple, can provide crucial data and functionality to other systems, especially in safety-critical contexts. Here's a breakdown of key functions and data, focusing on safety-critical outputs:

Key Functions/Data Provided by the Wiper System:

I. Safety-Critical Outputs (Directly impacting vehicle safety):

- **Wiper Status/State:** This is the most fundamental safety-critical output.

***Data:*

- **Wiper Active/Inactive:** Is the wiper motor currently running?
-

-
- **Wiper Speed/Mode:** What is the current speed setting (e.g., intermittent, low, high, auto)? This is crucial for the VCU to understand the level of visibility being provided.
 - **Wiper Position (Optional but highly beneficial):** Knowing if the wipers are at the park position or actively sweeping. This can be inferred from motor current or dedicated sensors.

***Why it's Safety-Critical:*

- **Visibility Assurance:** The VCU needs to know if the wipers are functioning to ensure the driver has adequate visibility in rain, snow, or other precipitation.
- **Emergency Braking/ADAS Integration:** In severe weather, the VCU might need to adjust ADAS (Advanced Driver-Assistance Systems) behavior (e.g., reduce speed, increase following distance, disable certain features) based on wiper status.
- **Driver Awareness:** The VCU can alert the driver if wipers are not functioning when expected, preventing them from driving with impaired visibility.

***Wiper Motor Fault/Failure Detection:*

***Data:*

- **Motor Overcurrent/Overload:** Indicates a potential obstruction or mechanical issue.
- **Motor Stalled/Not Moving:** The motor is receiving power but not rotating.
- **Communication Error (if CAN-based):** Loss of communication with the wiper control module.
- **Sensor Failure (if applicable):** Failure of rain sensors or position sensors.

***Why it's Safety-Critical:*

- **Preventing Catastrophic Failure:** A stalled wiper motor can lead to a complete loss of visibility, which is a direct safety hazard.
- **Diagnostic Information:** Allows the VCU to log faults, inform the driver, and potentially trigger limp-home modes or disable certain features that rely on clear visibility.

***Rain Sensor Data (if integrated):*

***Data:*

- **Rain Intensity Level:** Quantified measurement of precipitation (e.g., light, moderate, heavy).
 - **Rain Detected/Not Detected:** A binary indication.
-

***Why it's Safety-Critical:*

- **Automatic Wiper Control:** Enables the system to automatically adjust wiper speed based on the severity of the rain, ensuring optimal visibility without driver intervention.
- **Proactive ADAS Adjustment:** The VCU can use this data to proactively adjust ADAS parameters before visibility becomes critically impaired.

II. Non-Safety-Critical but Important Outputs (Enhancing user experience and system integration):

***Wiper Wash Function Status:*

***Data:*

- **Washer Pump Active/Inactive:** Is the washer fluid being dispensed?
- **Washer Fluid Level (if sensor present):** Indicates when the fluid is low.

***Why it's Important:*

- **Driver Feedback:** The VCU can provide feedback to the driver about the washer system's operation.
- **Preventing Dry Wiping:** In some systems, the VCU might prevent dry wiping if the washer fluid level is critically low.

***Wiper Motor Current/Load (Detailed):*

- **Data:** Precise measurement of the current drawn by the wiper motor.

***Why it's Important:*

- **Advanced Diagnostics:** Beyond simple fault detection, this can help diagnose the health of the motor and identify gradual wear or impending failures.
- **Power Management:** The VCU can use this information for more sophisticated power management strategies.

***Wiper Park Position Confirmation:*

- **Data:** A signal confirming that the wipers have returned to their designated park position.

***Why it's Important:*

- **System Integrity:** Ensures the wipers are stowed correctly, preventing them from obstructing the driver's view or being damaged.
- **User Interface:** Can be used to update the UI to reflect the wipers being parked.

III. Data Provided *to* the Wiper System (from SoC/VCU):

While the focus is on outputs *from* the wiper system, it's important to note that the wiper system also *receives* commands and data:

- **Wiper Command (Speed/Mode Selection):** From the driver's input (stalk, touchscreen) or from the VCU (e.g., automatic mode based on rain sensor).
- **Vehicle Speed:** The VCU might adjust wiper speed based on vehicle speed (e.g., faster wipers at higher speeds).
- **Ignition Status:** To enable/disable wiper operation.
- **System Enable/Disable Signals:** For specific driving modes or safety conditions.

In summary, the most critical safety-related data provided by the Wiper System to other systems like the SoC and VCU revolves around:

***Ensuring the driver has adequate visibility by reporting the active status and speed of the wipers.*

***Detecting and reporting any faults or failures that could compromise visibility.*

***Providing real-time environmental data (rain intensity) to enable proactive system adjustments.*

This information allows the VCU to make informed decisions about vehicle operation, ADAS functionality, and driver alerts, ultimately contributing to a safer driving experience.

5.4 Functionality of Other Items Required by This Item

Reference: 5.4.2 d)

The Wiper System, while seemingly simple, relies on several critical inputs from other vehicle systems to operate safely and effectively. Here's a breakdown of key functions/data required and their dependencies:

Key Functions/Data Required by the Wiper System:

1. VCU (Vehicle Control Unit) Mode Signal:

***Function/Data:** **Vehicle State/Mode Information*

***Specific Data:*

-
- **Ignition Status (ON/OFF):** Crucial for enabling or disabling wiper operation. Wipers should generally only operate when the ignition is ON.

***Vehicle Speed (Low/High/Parked):*

- **Low Speed/Parked:** May trigger a slower wiper speed or prevent operation altogether to avoid damage or unnecessary wear.
- **High Speed:** May require a faster wiper speed for effective clearing.

***Gear Selection (Park/Drive/Reverse):*

- **Park:** May influence wiper behavior (e.g., returning to park position).
- **Reverse:** Often triggers a higher wiper speed or intermittent operation to improve visibility when backing up.
- **Vehicle Operational State (e.g., Active, Standby, Fault):** The wiper system should be disabled or operate in a safe mode if the vehicle is in a critical fault state.
- **Critical Dependency: High.** Incorrect or missing VCU mode signals can lead to:
 - Wipers operating when the vehicle is off (battery drain).
 - Wipers not operating when needed (safety hazard).
 - Wipers operating at inappropriate speeds for the vehicle's state (damage, reduced effectiveness).
- Wipers interfering with other vehicle functions.

2. BCM (Body Control Module) / Lighting Control Module:

***Function/Data:** **Rain Sensor Input*

***Specific Data:*

- **Rain Intensity (None, Light, Medium, Heavy):** This is the primary input for automatic wiper operation.
- **Rain Sensor Status (Operational/Fault):** The BCM typically processes the raw sensor data and provides a simplified output to the wiper module.
- **Critical Dependency: High (for automatic wipers).** Without this, automatic wiper functionality is impossible. If the sensor fails, the system should revert to manual control or a default intermittent setting.

3. Body Control Module (BCM) / User Interface Module:

***Function/Data:** **User Input / Wiper Control Commands*

***Specific Data:*

- **Wiper Switch Position (Off, Intermittent (various speeds), Low Speed, High Speed, Wash/Wipe):** This is the direct command from the driver.
- **Wiper Speed Selection (for intermittent mode):** The driver's preference for how often the wipers sweep.
- **Critical Dependency: High.** This is the primary way the driver interacts with the wiper system. Any failure in transmitting these commands will render the wipers unusable or uncontrollable.

4. Powertrain Control Module (PCM) / Engine Control Module (ECM):

***Function/Data:** **Engine Status (e.g., Engine Running/Stopped)*

***Specific Data:*

- **Engine RPM (less critical, but can be used for advanced features):** In some advanced systems, engine load or RPM might influence wiper speed in conjunction with vehicle speed.
- **Critical Dependency: Medium.** While not as critical as VCU mode, knowing if the engine is running can be used to prevent wipers from operating when the vehicle is parked and the engine is off, but the ignition is on (e.g., for cleaning).

5. Gateway Module / Network Management:

***Function/Data:** **Communication Integrity and Message Routing*

***Specific Data:*

- **CAN Bus Status:** The wiper module relies on the vehicle's communication network (e.g., CAN bus) to receive all the aforementioned data.
- **Message Timeliness and Validity:** The gateway ensures that messages from other ECUs are received on time and are not corrupted.
- **Critical Dependency: High.** If the communication network fails or messages are lost/corrupted, the wiper system will not receive the necessary inputs and will cease to function correctly.

6. Actuator Control Module (often integrated within the Wiper Module itself):

***Function/Data:** **Motor Control Signals*

***Specific Data:*

-
- **Motor Speed Commands (PWM signals):** To control the speed of the wiper motor.
 - **Motor Direction Commands:** To ensure the wipers sweep in the correct direction.
 - **Motor Park Position Feedback:** To know when the wipers have returned to their resting position.
 - **Critical Dependency: High.** This is the direct output from the wiper control logic to the physical wiper motor.

7. Power Management Module / Battery Management System (BMS):

***Function/Data:** **Power Availability and Voltage Status*

***Specific Data:*

- **Battery Voltage:** The wiper system needs sufficient voltage to operate. In low voltage conditions, it might be disabled to conserve battery power.
- **Power Supply Status:** Ensuring a stable and reliable power source.
- **Critical Dependency: High.** Without adequate power, the wiper system cannot function.

8. Lighting Control Module (for Headlight/Wiper Integration):

***Function/Data:** **Headlight Status*

***Specific Data:*

- **Headlights ON/OFF:** Many vehicles automatically activate the wipers at a low intermittent speed when the headlights are turned on, especially in rain.
- **Critical Dependency: Medium (for automatic activation).** This is a convenience and safety feature, but not essential for basic wiper operation.

Summary of Critical Dependencies:

The most critical dependencies for the Wiper System are:

- **VCU Mode Signal:** Essential for safe and context-aware operation.
- **User Input (Wiper Switch):** The primary driver control.
- **Rain Sensor Input (for automatic wipers):** Enables automated functionality.
- **Gateway Module / Network Integrity:** The backbone of communication.
- **Power Supply:** Fundamental for any electrical system.

Failure in any of these critical areas can lead to a significant degradation in safety and functionality of the wiper system.

5.5 Allocation and Distribution of Functions

Reference: 5.4.2 e)

Here's how safety-critical functions are distributed in a wiper system and its interactions:

- **Wiper Motor Control & Fail-Safe:** The primary safety-critical function is the reliable operation of the wiper motor to clear the driver's view. This is typically managed by a dedicated Electronic Control Unit (ECU) or integrated into a Body Control Module (BCM). This ECU monitors for faults (e.g., motor stall, electrical issues) and can implement fail-safe mechanisms like stopping the wipers in a non-obstructive position or defaulting to a slow intermittent speed if a critical failure is detected.
- **Integration with Other Vehicle Systems (e.g., Rain Sensor, Lighting, ADAS):** Safety-critical aspects extend to how the wiper system interacts with other vehicle systems. A rain sensor's input is crucial for automatic wiper activation, and its failure might trigger a default manual mode or a driver alert. Furthermore, in advanced systems, wiper operation might be linked to Automatic Emergency Braking (AEB) or Adaptive Cruise Control (ACC) logic, where impaired visibility due to ineffective wipers could influence the activation or deactivation of these ADAS features, requiring coordinated safety responses.

5.6 Operational Scenarios Impacting Functionality

Reference: 5.4.2 f)

Here are a few key operational scenarios that significantly impact the 'Wiper System' safety considerations, informing Hazard Analysis and Risk Assessment (HARA):

1. Sudden Obstruction/Jamming:

- **Scenario:** The wiper blades encounter a significant obstruction that prevents them from moving freely. This could be:
- **Ice/Snow Buildup:** Heavy accumulation of ice or snow on the windshield or the wiper blades themselves.
- **Debris:** A large leaf, branch, or other foreign object getting caught under the wiper arm.
- **Mechanical Failure:** A component within the wiper mechanism (e.g., linkage, motor) seizes or breaks.

***Safety Impact:*

-
- **Motor Overheating/Failure:** The wiper motor will strain against the obstruction, leading to overheating and potential burnout. This renders the system inoperable.
 - **Damage to Wiper Arms/Blades:** Excessive force can bend or break wiper arms, or damage the blades, making them ineffective.
 - **Potential for Injury:** If the obstruction is sudden and the wiper arm is forcefully stopped, it could potentially cause minor injury to a passenger if they are near the windshield or if the arm snaps back.
 - **Loss of Visibility:** The primary safety concern is the immediate loss of the ability to clear the windshield, leading to reduced visibility.

2. Sudden Loss of Wiper Functionality (Electrical/Mechanical Failure):

- **Scenario:** The wiper system stops working unexpectedly during operation. This could be due to:
- **Fuse Blown/Circuit Breaker Tripped:** An electrical fault causes a protective device to interrupt power.
- **Motor Failure:** The wiper motor itself fails internally.
- **Wiring Harness Damage:** A break or short in the wiring connecting the motor to the power source.
- **Relay Failure:** The relay controlling the wiper motor malfunctions.

***Safety Impact:*

- **Sudden Loss of Visibility:** This is the most critical safety impact. If the wipers stop mid-wipe during heavy rain or snow, the driver is suddenly left with a partially or completely obscured windshield.
- **Driver Distraction:** The driver may be distracted trying to diagnose or fix the issue, taking their attention away from the road.
- **Increased Risk of Collision:** Reduced visibility significantly increases the risk of colliding with other vehicles, pedestrians, or obstacles.

3. Wiper Blade Detachment/Separation:

- **Scenario:** One or both wiper blades detach from the wiper arm during operation. This can happen due to:
 - **Improper Installation:** The blade not being securely attached to the arm.
 - **Wear and Tear:** The connection mechanism on the blade or arm becoming worn and failing.
-

-
- **High Wind/Aerodynamic Forces:** In extreme conditions, the blade can be lifted and detached.

***Safety Impact:*

- **Sudden Loss of Visibility:** The detached blade can no longer clear the windshield, leading to immediate visibility issues.
- **Projectile Hazard:** A detached wiper blade can become a projectile, posing a risk to other vehicles, pedestrians, or even the occupants of the same vehicle if it strikes the windshield with force.
- **Damage to Windshield:** A detached blade can scratch or crack the windshield.
- **Driver Distraction:** The driver's attention will be diverted by the sudden event and the potential hazard.

4. Inadvertent Activation/Operation:

- **Scenario:** The wiper system activates or operates when it is not intended to. This could be due to:
- **Faulty Switch/Stalk:** The wiper control switch on the steering column malfunctions.
- **Electrical Cross-Talk:** An unintended electrical signal triggers the wiper system.
- **Software Glitch (for intelligent systems):** A bug in the vehicle's software causes the wipers to activate.

***Safety Impact:*

- **Distraction:** The unexpected movement of the wipers can be distracting to the driver, especially if they are not expecting it.
- **Reduced Visibility (if activated inappropriately):** If the wipers are activated on a dry windshield, they can cause smearing and reduce visibility. If they are activated at a high speed on a dry surface, they can cause damage to the blades and windshield.
- **Potential for Damage:** Operating wipers on a dry windshield can cause wear and tear on the blades and scratch the glass.

5. Inadequate Wiper Performance (Low Speed/Ineffective Clearing):

- **Scenario:** The wiper blades move, but at an insufficient speed or with insufficient pressure to effectively clear water, snow, or debris from the windshield. This can be caused by:
 - **Weak Motor:** The wiper motor is losing power.
-

-
- **Worn Wiper Blades:** The rubber on the blades is degraded and no longer makes good contact with the windshield.
 - **Bent Wiper Arms:** The arms are not applying enough pressure to the windshield.
 - **Low Washer Fluid Pressure:** If the system is designed to work in conjunction with the washer fluid, low pressure can lead to ineffective clearing.

***Safety Impact:*

- **Gradual Reduction in Visibility:** The driver's view of the road is progressively impaired, leading to a higher risk of misjudging distances or failing to see hazards.
- **Increased Driver Fatigue:** The driver may have to strain their eyes more to see, leading to fatigue.
- **Complacency:** The driver might not immediately recognize the severity of the reduced visibility, leading to a delayed reaction to hazards.

These scenarios are crucial for HARA as they highlight potential hazards that could lead to hazardous events and ultimately, safety goals being violated. For each scenario, a HARA would then identify the specific hazards, their causes, potential consequences, and the necessary safety measures to mitigate the risks.

6. Open Points

Reference: 5.4.7

Here are essential unresolved issues and critical assumptions for a "Wiper System":

Unresolved Issues:

- **Durability/Longevity:** How long will the wiper blades and motor last under various environmental conditions (UV, extreme temperatures, debris)? What is the expected failure rate?
 - **Performance in Extreme Conditions:** How effectively will the system clear heavy rain, snow, ice, or mud? Will it maintain consistent contact with the windshield?
 - **Power Consumption/Efficiency:** What is the power draw of the motor, especially during intermittent or high-speed operation? Is it optimized for battery life (if applicable)?
 - **Noise/Vibration:** What are the acceptable noise and vibration levels during operation? Are there potential sources of squeaking, rattling, or motor hum?
-

-
- **Maintenance/Replacement:** How easy is it for the end-user to replace wiper blades? What are the recommended maintenance intervals for the motor and linkage?
 - **Corrosion Resistance:** Are all components adequately protected against rust and corrosion, especially in environments with salt or high humidity?
 - **Obstruction Handling:** How will the system react if the wiper blades encounter an obstruction (e.g., a fallen branch, ice buildup)? Will it stall, reset, or cause damage?
 - **Integration with Other Systems:** If part of a larger vehicle, how does it interact with rain sensors, defogging systems, or autonomous driving features? Are there potential conflicts?

Critical Assumptions:

- **Windshield Design:** The windshield shape and curvature are suitable for the chosen wiper blade and arm design to achieve effective coverage.
- **Power Availability:** A reliable and sufficient power source is available to operate the wiper motor at the required speeds and durations.
- **Environmental Tolerance:** The system is designed to operate within a specified range of ambient temperatures, humidity, and exposure to common environmental contaminants.
- **User Interaction:** Users will operate the system as intended and understand basic maintenance procedures (e.g., blade replacement).
- **Material Properties:** The materials used for blades, arms, and motor components will maintain their intended mechanical properties (flexibility, strength, resistance to wear) for the expected lifespan.
- **Mounting Stability:** The wiper arm and motor assembly will remain securely mounted to the vehicle/structure without loosening or shifting over time.
- **Fluid Delivery (if applicable):** If a washer fluid system is integrated, it will reliably deliver fluid to the windshield at the required pressure and volume.

7. Proven in Use Argument

Reference: ISO 26262-8:2018, Clause 12

Wiper System: Proven in Use

Components:

- **Wiper Blades:** Rubber or silicone elements that contact the windshield.
 - **Wiper Arms:** Metal or composite structures that hold the blades and pivot.
-

-
- **Wiper Motor:** Electric motor that drives the wiper arms.
 - **Wiper Linkage:** Mechanical system connecting the motor to the arms.
 - **Control Module/Switch:** Electrical component for user operation and speed selection.

Field Data:

- **Reliability:** Decades of widespread use demonstrate high reliability in various environmental conditions (rain, snow, dust).
- **Durability:** Blades are designed for seasonal replacement; arms and motors have lifespans typically exceeding 100,000 cycles.
- **Performance:** Effective removal of precipitation and debris for visibility.
- **Maintenance:** Routine blade replacement is standard; occasional lubrication of linkage points.
- **Failure Modes:** Blade degradation (cracking, tearing), motor burnout, linkage wear, electrical connection issues. These are well-understood and addressed through design and maintenance.

8. SEooC Development Context

Reference: ISO 26262-8:2018, Clause 13

Integration Environment:

- **Hardware:** Dedicated safety controller (e.g., automotive-grade microcontroller), sensors (rain sensor, wiper motor position feedback), actuators (wiper motor driver), communication interfaces (CAN, LIN).
- **Software:** Real-time operating system (RTOS), safety-certified middleware, diagnostic software, communication stacks.
- **Vehicle System:** Integration with vehicle power management, ignition status, lighting systems, and potentially advanced driver-assistance systems (ADAS) for context-aware operation.

Assumptions:

- **Defined Safety Goals:** Clear and verifiable safety goals for the wiper system (e.g., prevent obstruction of driver's view in critical conditions).
 - **Hardware Reliability:** The underlying hardware components (controller, sensors, actuators) meet specified reliability and safety integrity levels (ASIL).
 - **Communication Integrity:** Communication channels are sufficiently robust and protected against errors and interference.
-

-
- **Environmental Conditions:** The system is designed to operate within specified environmental ranges (temperature, humidity, vibration).
 - **Power Supply Stability:** A stable and reliable power supply is available.
 - **Diagnostic Capabilities:** The integration environment provides mechanisms for fault detection, reporting, and potentially mitigation.

9. Component Reuse

Reference: ISO 26262-8:2018, Clause 14

To identify reused components for a "Wiper System," I need more context. Please provide information about:

- **The specific wiper system you are referring to:** Is this for a particular vehicle model, a new product development, a simulation, or something else?
- **The source of information:** Where did you get the description or specifications of this wiper system?

Without this information, I cannot definitively identify reused components.

However, I can provide a general framework for how this would be assessed and what common reused components might be found in a wiper system:

General Approach to Identifying Reused Components:

1. **Component Breakdown:** The first step is to have a detailed list of all components within the wiper system. This would include:

- Motor
- Gearbox/Reduction mechanism
- Linkage (wiper arms, pivot points)
- Wiper blades
- Control module/ECU
- Wiring harness
- Sensors (rain sensor, park sensor)
- Mounting hardware

2. **Comparison with Existing Designs:** Each component would then be compared against:

- **Previous versions of the same wiper system:** If this is an upgrade or refresh.

-
- **Other wiper systems from the same manufacturer:** For cost-saving and standardization.
 - **Industry-standard components:** Many components, especially motors and basic sensors, are sourced from specialized suppliers and are used across multiple applications.
3. **Justification for Reuse:** If a component is identified as reused, the justification would typically be:
- **Cost Reduction:** Leveraging existing tooling, manufacturing processes, and bulk purchasing.
 - **Proven Reliability:** Using components that have a track record of performance and durability.
 - **Faster Development Time:** Avoiding the need for new design, prototyping, and testing.
 - **Standardization:** Maintaining consistency across product lines.
 - **Supplier Relationships:** Continuing to work with established and trusted suppliers.

Potential Reused Components in a Wiper System (Examples):

If this were a typical automotive wiper system, here are some components that are **highly likely** to be reused:

- **Wiper Motor:** Automotive manufacturers often use a limited range of wiper motors across different vehicle platforms to achieve economies of scale.
 - **Justification:** Cost reduction, proven reliability, established supplier relationships.
 - **Gearbox/Reduction Mechanism:** The internal gearing that translates the motor's rotation into the sweeping motion of the wiper arms is often standardized.
 - **Justification:** Cost reduction, proven reliability, standardization.
 - **Wiper Blades (the rubber element):** While the arms might be specific, the actual rubber wiping element is often a standard part sourced from specialized suppliers.
 - **Justification:** Cost reduction, proven reliability, standardization.
 - **Basic Sensors (e.g., Park Sensor):** The sensor that tells the motor when to stop in the park position is often a simple, robust, and standardized component.
 - **Justification:** Cost reduction, proven reliability, standardization.
 - **Wiring Harness Connectors/Terminals:** Standard electrical connectors and terminals are almost always reused.
 - **Justification:** Cost reduction, standardization.
-

If you can provide more details about the specific wiper system, I can give you a more precise answer.

11. Abbreviations and References

Reference: General Documentation

Please provide the document you would like me to analyze. I need the text of the document to identify abbreviations, standards, regulations, and internal documents.

12. Conclusion

Reference: 5.4.7

This document satisfies ISO 26262-3:2018 Clause 5.4 by providing the necessary inputs for Hazard Analysis and Risk Assessment (HARA) and the subsequent development of the Safety Concept.

13. Approvals

Reference: ISO 26262-8:2018, Clause 6

Document Approval

This document requires approval from the following key stakeholders to signify their agreement with its contents and their commitment to its implementation.

Role	Name	Signature	Date
------	------	-----------	------

-----	-----	-----	-----
-------	-------	-------	-------

/ FSM (Functional Safety Manager)	/ [Placeholder for FSM Name]	/ [Placeholder for FSM Signature]	/ [Placeholder for Date]
-----------------------------------	------------------------------	-----------------------------------	--------------------------

/ SA (System Architect)	/ [Placeholder for SA Name]	/ [Placeholder for SA Signature]	/ [Placeholder for Date]
-------------------------	-----------------------------	----------------------------------	--------------------------

/ PM (Project Manager)	/ [Placeholder for PM Name]	/ [Placeholder for PM Signature]	/ [Placeholder for Date]
------------------------	-----------------------------	----------------------------------	--------------------------

/ QA (Quality Assurance)	/ [Placeholder for QA Name]	/ [Placeholder for QA Signature]	/ [Placeholder for Date]
--------------------------	-----------------------------	----------------------------------	--------------------------